

Pilot Test Report and Internal Validation

Project: Sahayak - Welfare Scheme Discovery Assistant
Official expected deliverable: Pilot test report with 10-15 real users from any nearby village or community

Evidence Status

This report is an honest internal validation of the current prototype. It uses scripted beneficiary personas, automated tests, and manual walkthroughs completed by the project team on 13 June 2026.

It does not claim feedback from 10-15 external community users. If a field pilot or short demo is completed before final submission, the real-user appendix at the end of this document can be filled in directly.

Validation Objective

- Evaluate whether Sahayak can help a user:
- 1. Enter an eligibility profile in English or Hindi.
 - 2. Receive relevant welfare scheme recommendations.
 - 3. Understand why a scheme was suggested.
 - 4. Ask a follow-up question.
 - 5. Prepare a document checklist for application.
 - 6. Verify the result through official source links.

Test Setup

Field	Details
Date	13 June 2026
Tester type	Internal project validation
Prototype	Static GitHub Pages web app
Live URL	https://parzival1821.github.io/Multilingual_Chatbot/
Repository	https://github.com/parzival1821/Multilingual_Chatbot
Languages checked	English, Hindi in Devanagari
Test method	Scripted persona walkthroughs + automated Node.js checks
Evidence files	tests/core.test.js, tests/static-smoke.test.js, README, submission docs

Scripted Persona Coverage

Persona	Key profile signals	Expected scheme
Landholding farmer	Rural, farmer, cultivable land, bank account	PM-KISAN, PM-J
Woman-led household	Female applicant, low income, no LPG, no permanent house, girl child guardian	Ujjwala, PMAY-U
Street vendor	Urban street vendor, low income, bank account	PM SVANidhi, PM
Widow aged 55	Widow, BPL/low income, social assistance need	NSAP
Health-document seeker	Asks for Ayushman Bharat documents	PM-JAY checklis
Unsupported query	Asks outside current knowledge base	Safe fallback with

Manual Walkthrough Results

Test task	Expected behavior
Open the live prototype	App loads without login and shows Sahayak dashboard
Switch to Hindi	UI copy, prompt examples, and scheme names use Devanagari
Use Farmer demo preset	Form fills a farmer profile without hiding editable fields

Test task	Expected behavior
Click Find Schemes	Recommendations are ranked with reasons and source links
Click View documents	Checklist panel updates to the selected scheme and scrolls into view
Ask What documents do I need for Ayushman Bharat?	Assistant returns PM-JAY document guidance with source link
Ask I am widow of age 55 which scheme is best for me?	Assistant routes to NSAP social assistance guidance
Copy checklist	Checklist text copies in a low-bandwidth-friendly format
Download checklist	Browser downloads a plain-text checklist
Ask unsupported startup subsidy query	Assistant avoids hallucination and asks user to verify official sources

Automated Validation

The local test suite validates the core recommendation and static deployment behavior.

Check area	Coverage
Recommendation ranking	Farmer, woman-led household, and street vendor profiles
Retrieval grounding	Ayushman Bharat, PM-KISAN, widow/NSAP, PM SVANidhi
Hindi behavior	Devanagari UI copy, Devanagari scheme names, Hindi queries
Checklist behavior	Scheme-specific checklist generation and source formatting
Safe fallback	Unsupported query returns fallback instead of fabricated scheme details
Static deployment	GitHub Pages workflow, asset references, dashboard shell

Expected local verification command:

```
npm run check
```

Expected result:

All core tests passed.

Static smoke checks passed.

Internal Metrics

These are prototype-validation metrics, not external community-pilot metrics.

Metric	Internal result	Notes
Scripted flow completion	6/6 scenarios passed	All scripted personas reached a useful result
Recommendation correctness proxy	6/6 scenarios passed	Based on expected scheme-persona mapping
Checklist availability	8/8 schemes covered	Every scheme has application documents
Source grounding	8/8 schemes covered	Every scheme has at least one official source link
Language coverage	2 languages	English and Hindi Devanagari
Response speed	Not formally instrumented	Static local/browser interactions appeared immediate during

Findings

1. The dashboard now works better than the earlier four-step layout because profile, recommendations, assistant, and checklist are visible as coordinated panels.
2. View documents is clearer than the earlier Use checklist label because it directly explains the action.
3. Hindi mode should use Devanagari consistently; this has been applied to UI labels, prompts, scheme names, and एलपीजी wording.
4. Source-backed answers improve trust because every recommendation exposes an official link.
5. The largest remaining validation gap is external field feedback from real beneficiaries or NSS/community volunteers.

Improvements Made After Validation

Issue noticed	Change made
Four-step page felt rough and beginner-like	Reorganized into a single dashboard with dedicated panels
Checklist action did not communicate behavior	Changed to View documents and connected it to checklist selection
Hindi mode used Hinglish in places	Replaced with Devanagari copy
Scheme names stayed in English in Hindi mode	Added Hindi display names for schemes
Local terminology like pucca was unclear in English	Reworded to permanent house

Real-User Pilot Appendix

Use this table only if a real demo or field test is completed. Do not fill it with assumed data.

User ID	Persona / background	Language	Completed flow?	Top recommendation understood?	Checklist useful?	N
U01	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U02	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U03	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U04	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U05	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U06	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U07	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U08	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U09	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U10	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U11	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U12	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U13	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U14	To be filled	To be filled	Yes/No	Yes/No	1-5	N
U15	To be filled	To be filled	Yes/No	Yes/No	1-5	N

Conclusion

The current prototype passes internal validation for the submission demo: it can recommend schemes, explain match reasons, answer grounded follow-up questions, and produce document checklists in English and Hindi.

For a stricter interpretation of the official pilot requirement, the remaining step is to run the same script with 10-15 real users or community representatives and paste observations into the appendix above.